

RAJ S

Nagarajan Shunmugam

AI Engineer · CRISPR & NGS Researcher · Co-Founder & CTO · Future Veterinarian (DVM Candidate)

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PROFILE

Nine years of applied AI across six countries: autonomous trucks at Scania, medical imaging in Berlin, AI for the world's largest veterinary community, and CRISPR machine learning in Wrocław. Three-time startup co-founder. Currently a CRISPR & NGS data researcher at the Hirsfeld Institute (Polish Academy of Sciences), CTO of MilkAI, and a veterinary medicine student, building the bridge between AI and animal health.

EXPERIENCE

- CRISPR & NGS Data Researcher** · Hirsfeld Institute of Immunology & Experimental Therapy, PAN
Wrocław, Poland
ML pipelines for CRISPR off-target prediction on genomic, NGS and multi-omics data (k-mer, scalar, one-hot encodings; CNN and Transformer models). Multi-agent AI systems that design and validate wet-lab experiment protocols.
Nov 2025 – present
- Co-Founder & CTO** · MilkAI LLC
California, USA
Product studio shipping production-ready web, mobile and AI-first apps, with design-to-deploy CI/CD. Dozens of launches.
Jul 2025 – present
- Artificial Intelligence Engineer** · Veterinary Information Network
Remote (US company)
AI engineering for the world's largest online veterinary community.
May 2023 – Jul 2025
- Co-Founder** · Zaide.ai
Remote
Built private, custom large language models for healthcare and life sciences: secure local LLMs designed so sensitive clinical and research data never leaves the organization.
Jan 2023 – Aug 2024
- Mentor & Alpha Tester** · Coursera
Orange County, CA
Six years mentoring the Neural Networks & Deep Learning course; alpha-tested unreleased AI courses; mentored 3D-printing applications.
Apr 2017 – Jun 2023
- Assistant Lecturer** · Coventry University (Wrocław campus)
Wrocław, Poland
Taught AI, IoT and software engineering.
Aug 2022 – Mar 2023
- Senior AI Instructor** · Techeta Technologies
India
Trained technical and corporate teams in applied AI.
Sep 2021 – Sep 2022
- Co-Founder & CTO** · Neurons4vet.ai
Toronto, Canada
First venture at the intersection of AI and veterinary medicine.
Jan 2020 – Dec 2021
- Cohort Member** · Entrepreneur First
Berlin, Germany
Europe's leading deep-tech talent investor (300+ startups, over \$2.7B portfolio).
Mar – Aug 2021
- Thesis Student** · Scania Group
Stockholm, Sweden
Truck visual perception: classification, detection, semantic segmentation, depth estimation, reinforcement learning.
Jan 2020 – Mar 2021
- Machine Learning Engineer** · ai4medicine
Berlin, Germany
May – Oct 2020

Computer-vision AI for stroke and bleeding detection.

Senior Design Engineer · KTH Formula Student

Aug 2019 – Jan 2020

Stockholm, Sweden

Autonomous race car perception; LiDAR point clouds; NVIDIA Jetson deployment.

Project Assistant · Indian Institute of Technology, Delhi

Sep 2017 – Feb 2018

New Delhi, India

Driverless car: PID tuning, GPS+IMU+LiDAR+radar sensor fusion.

Research Scholar · Indian Institute of Technology, Madras

Mar – Aug 2017

Chennai, India

Embedded programming, biomedical research lab (bio-medical kiosks, SpO2 sensors).

EDUCATION

Wrocław University of Environmental and Life Sciences (UPWR) · Veterinary

2021 – 2027

Medicine, Veterinary Surgeon (DVM), in progress

Technische Universität Berlin · M.Sc. Autonomous Systems, Robotics & AI (full scholarship)

2018 – 2021

KTH Royal Institute of Technology · M.Sc. Autonomous Systems, Robotics & AI (EIT Digital double degree)

2018 – 2021

Aalto University · EIT Digital Summer School: Disrupting Retail

2019

Anna University · B.E. Electrical & Electronics Engineering (partial scholarship)

2013 – 2017

SKILLS

AI/ML: CNNs & Transformers, deep reinforcement learning, imitation learning, multi-agent systems, data analysis ·

Genomics: CRISPR guide design, NGS & multi-omics, k-mer/one-hot encodings · **Autonomy:** detection & segmentation, depth estimation, LiDAR, sensor fusion, NVIDIA Jetson · **Engineering:** technical leadership, IoT & embedded (Arduino), 3D printing, CI/CD

PUBLICATION & HONORS

Publication: Deep Learning for Hardware-Constrained Cars: Imitation Learning & Deep Reinforcement Learning

Honors: Full scholarship (master's) · Partial scholarship (bachelor's) · **Languages:** Bilingual, English (IELTS C1) and Tamil